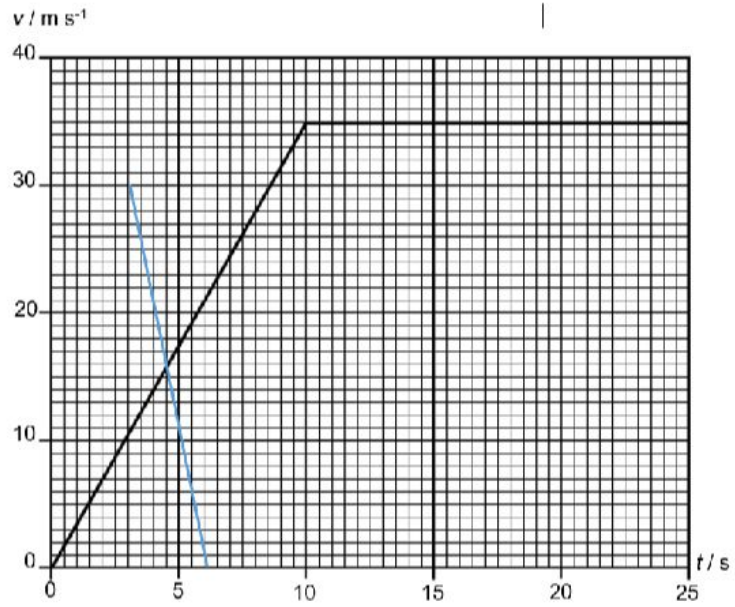
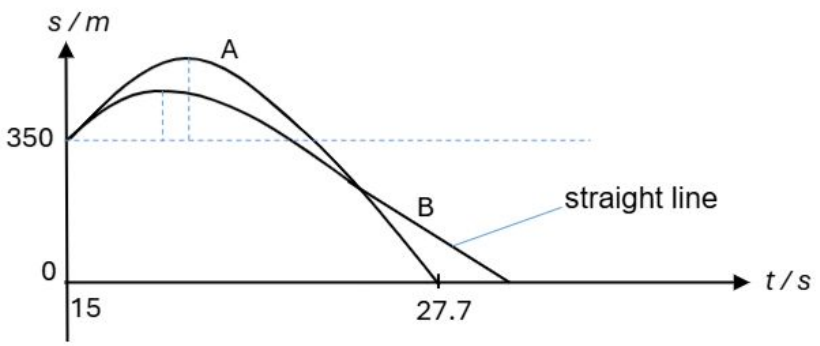
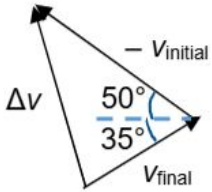
	$t = 9.9 \times 10^{7}$ years	A1
8a	Gradient = (35.0 – 0.0) / (10.0) = 3.50 m s^{-2}	A1
8bi	A straight line graph with gradient about – 10, v = 30 m s^{-1} at t = 3.0 s, v = 0 at t = 6.0 s. (see 8bii for the graph)	A1
8bii	Min separation occurs when velocity of phone v_p = velocity of boy (balloon) v_b Intersection at t ~ 4.5 s  Separation = distance by boy – distance by phone = ½(3.50)(4.5)^2 – [(30)(1.5) – ½ (9.81)(1.5)^2] = 1.5 m OR separation = distance by boy – distance by phone = ½(4.5)(16.0) – ½(1.5)(30+16.0) (Areas estimated from graphs) = 1.5 m (accept 0.75 to 1.5 m) OR velocity of phone = velocity of boy	M1 C1 A1 (M1)

	$u_p - g(t-3.0) = u_b + at$ $30 - (9.81)(t-3.0) = 0 + (3.50)t$ t = 4.465 = 4.47 s Separation = distance by boy – distance by phone = ½ (3.50)(4.47)^2 – [(30)(1.47) – ½ (9.81)(1.47)^2] = 1.5 m	(C)1 (A1)
8ci	Taking downwards as positive, $s_y = u_y t + \frac{1}{2} g t^2$ 0.5(35)(5+15) = − 35t + ½ (9.81)t^2 ➔ t = 12.74 = 12.7 s	C1 A1
8cii & ciii	 c)ii) smooth (parabolic) curve <u>symmetrical</u> about the highest point t = 15 s, s = 350 m & t = 27.7 s, s = 0 m **c)iii)** Initially a curve with the <u>same angle</u> as A but reach a <u>lower max</u> height at <u>an earlier time</u> than A and <u>asymmetry</u> about the highest point must be correct (longer time to return to the same height). Final section of the sketch is a straight line graph Same initial s & reaches ground later than that in (b)(ii)	B1 B1 B1 B1 B1
8di	Since the floor is frictionless, there is <u>no horizontal force</u>. Therefore, linear <u>momentum</u> is <u>conserved</u> in the <u>horizontal</u> direction. COLM in horizontal direction m (11.5 cos 50°) = m v cos 35° v = 9.02 = 9.0 m s^{-1} **OR** Since the floor is frictionless, there is <u>no horizontal force</u>. Therefore, there is <u>no horizontal acceleration</u> and <u>horizontal velocity</u> remains <u>unchanged</u>. 11.5 cos 50° = v cos 35° v = 9.02 = 9.0 m s^{-1}	M1 A1 M1 A1

8dii	$\Delta v = \sqrt{(11.5)^2 + (9.02)^2 - 2(11.5)(9.02)\cos 85°}$ $= 13.98 = 14$ m s^{-1}. Diagram for reference only:  	M1 A1
8diii	Let F be the contact force from the ground on the ball. $F - mg = m\Delta v/\Delta t$ $F = (0.500)(13.98)/(0.20) + (0.500)(9.81)$ $= 40$ N	 C1 A1
8div	There is work done against friction in horizontal direction, (so some kinetic energy is transferred to internal store). The ball will rebound with a smaller (horizontal) speed.	B1 B1
9a(i)	$R = \frac{\rho l}{A}$	